ADS401 EVALUATION METHODS AND STATISTICS

Usability Evaluation of the XJTLU Museum AI System

Task success, usability, and visitor barriers in a short museum session

PRESENTATION 2

Research design and pilot plan

GROUP H

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8-minute presentation

A public-facing AI guide must work without a manual

Practical problem

Visitors may not know how to start the guide, what to ask, where to go next, or how to recover from an unclear response.

The Museum needs direct evidence about the actions visitors actually perform.

RQ2 Can visitors complete the four core Museum AI tasks?
How usable and efficient is the system according to task performance and SUS?

RQ3 Which onboarding, navigation, answer, or recovery barriers recur?

TARGET USERS XJTLU students and staff, 18+, mixed digital confidence and museum experience

Four tasks expose the full visitor journey

TASK 01

Start

Open the Museum AI guide and select an exhibit.

TASK 02

Ask

Ask one exhibit question and identify the answer.

TASK 03

Navigate

Move to the next recommended exhibit or stop.

TASK 04

Recover

Respond when an answer is unclear or unsuitable.

METHOD

Moderated task-based usability test with think-aloud comments.

SAMPLE

12 to 15 participants, following the module usability guidance.

RATIONALE

Direct observation shows where users hesitate, fail, or need assistance.

A 15-minute session with one moderator

1. **Consent**
   Information sheet and agreement
2. **Warm-up**
   Museum and system context
3. **Tasks**
   Four observed scenarios
4. **SUS**
   Ten quick ratings
5. **Debrief**
   Barriers and improvement ideas

- Information sheet and consent form
- Moderator script and task cards
- Observation sheet and timer
- SUS and exit questions

Safety: no passwords, photographs, personal museum records, or sensitive university information.

Participants may stop at any time.

Observed performance and self-report answer different questions

Measure	What we record	How we analyse it
Task success	Independent, assisted, or failed	Success rate by task
Time on task	Seconds from instruction to outcome	Median and range
Errors and help	Wrong action, repeat attempt, moderator help	Frequency by task
SUS	Ten-item usability score	Mean score and benchmark comparison
Ease and trust	1-7 ratings after the tasks	Descriptive comparison
Think-aloud	Hesitation, confusion, workaround	Short theme summary

H1

At least 80% of participants complete each task without critical failure.

H2

The median SUS meets the accepted usability benchmark.

H3

Onboarding, navigation, or answer clarity dominate the barriers.

The study package is ready for a small pilot

READY

Participant materials

- Information sheet
- Consent form
- Moderator script
- Four task cards

CHECK

System access

- Confirm the current AI version
- Prepare a test account if required
- Prepare a fallback walkthrough

NEXT

Pilot and main sessions

- Pilot with two participants
- Revise unclear tasks
- Run 12-15 main sessions

DATA STATUS No participant data yet. We present the design and implementation plan, not invented findings.

The output is a concrete usability diagnosis

01 Task success profile for starting, asking, navigating, and recovering.

02 Evidence about where time, errors, or moderator help increase.

03 A usable SUS score with perceived ease and trust.

04 Prioritised recommendations for the Museum AI interface.

Q&A

Thank you. We welcome questions about the task design, sampling, measures, and implementation plan.